

6-1 · Division Expressions with Fractions and Whole Numbers

Connect & apply · 8 tasks · Prepares for 6.NS.A.1

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Compute $8.4 \div 4$.

2.1

84 tenths divided by 4 is 21 tenths.

Skill: Divide decimals and whole numbers

2. FOUNDATION 2

What is the reciprocal of 5?

$\frac{1}{5}$

Write 5 as $\frac{5}{1}$, then exchange numerator and denominator.

Skill: Name the reciprocal of a whole number and of a unit fraction

3. FOUNDATION 3

Compute $12 \times \frac{2}{3}$.

8

Divide 12 by 3, then multiply by 2.

Skill: Multiply a whole number by a fraction

4. FOUNDATION 4

Write $\frac{15}{3}$ as a division expression and evaluate it.

$15 \div 3 = 5$

A fraction bar means divide the numerator by the denominator.

Skill: Read a fraction bar as "divided by"

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5. CHECK A

$4.8 \div 2$? How many fourths are in 3? Reciprocal of 5?

$4.8 \div 2 = 2.4$. There are 12 fourths in 3 (since $3 \div 1/4 = 3 \times 4 = 12$). Reciprocal of 5 is $1/5$.

Compare each part of your work with the worked solution.

6. CHECK B

$6.4 \div 2$? How many thirds are in 4? Reciprocal of 8?

$6.4 \div 2 = 3.2$. There are 12 thirds in 4 ($4 \div 1/3 = 4 \times 3$). Reciprocal of 8 is $1/8$.

Compare each part of your work with the worked solution.

7. CHECK C

$9.6 \div 4$? How many halves are in 5? Reciprocal of $1/6$?

$9.6 \div 4 = 2.4$. There are 10 halves in 5. Reciprocal of $1/6$ is 6.

Compare each part of your work with the worked solution.

8. TRANSFER

A ribbon is 6 feet long. How many $3/4$ -foot pieces can be cut from it?

$6 \div 3/4 = 6 \times 4/3 = 24/3 = 8$ pieces.

Explain why your method works and include units when needed.

6-2 · Division Expressions with Fractions and Mixed Numbers

Connect & apply · 8 tasks · Prepares for 6.NS.A.1

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Write $2\frac{3}{4}$ as an improper fraction.

$$\frac{11}{4}$$

Two wholes contain 8 fourths; add 3 fourths to get 11 fourths.

Skill: Rewrite a mixed number as an improper fraction

2. FOUNDATION 2

Compute $\frac{2}{3} \times \frac{3}{5}$.

$$\frac{2}{5}$$

Multiply to get $\frac{6}{15}$, then simplify by dividing both parts by 3.

Skill: Multiply two fractions

3. FOUNDATION 3

Find the reciprocal of $\frac{4}{7}$.

$$\frac{7}{4}$$

Exchange numerator and denominator; their product will be 1.

Skill: Find the reciprocal of any fraction

4. FOUNDATION 4

Simplify $\frac{18}{24}$.

$$\frac{3}{4}$$

Divide numerator and denominator by their greatest common factor, 6.

Skill: Simplify a fraction to lowest terms

6-2 · Division Expressions with Fractions and Mixed Numbers

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Use the explanation to discuss strategies. Accept mathematically equivalent answers.

5. CHECK A

$2\frac{3}{4}$ as an improper fraction? Reciprocal of $\frac{3}{5}$?
 $\frac{2}{3} \times \frac{5}{4}$?

$2\frac{3}{4}$ as improper fraction: $(2 \times 4 + 3) / 4 = 11/4$. Reciprocal of $\frac{3}{5}$ is $\frac{5}{3}$. Product: $\frac{2}{3} \times \frac{5}{4} = \frac{10}{12} = \frac{5}{6}$.

Compare each part of your work with the worked solution.

6. CHECK B

$3\frac{2}{5}$ as an improper fraction? Reciprocal of $\frac{4}{7}$?
 $\frac{3}{4} \times \frac{2}{5}$?

$3\frac{2}{5} = (3 \times 5 + 2)/5 = 17/5$. Reciprocal of $\frac{4}{7}$ is $\frac{7}{4}$. $\frac{3}{4} \times \frac{2}{5} = \frac{6}{20} = \frac{3}{10}$.

Compare each part of your work with the worked solution.

7. CHECK C

$1\frac{5}{6}$ as an improper fraction? Reciprocal of $\frac{9}{2}$?
 $\frac{5}{6} \times \frac{3}{10}$?

$1\frac{5}{6} = \frac{11}{6}$. Reciprocal of $\frac{9}{2}$ is $\frac{2}{9}$. $\frac{5}{6} \times \frac{3}{10} = \frac{15}{60} = \frac{1}{4}$.

Compare each part of your work with the worked solution.

8. TRANSFER

How many $1\frac{1}{2}$ -cup servings are in 6 cups?

$6 \div \frac{3}{2} = 6 \times \frac{2}{3} = 4$ servings.

Explain why your method works and include units when needed.

6-3 · Explore Numerical Expressions with Exponents

Connect & apply · 8 tasks · Prepares for 6.EE.A.1

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Compute $3 \times 3 \times 3 \times 3$.

81

Successive products are 9, 27, and 81.

Skill: Multiply the same factor repeatedly

2. FOUNDATION 2

Evaluate 5^2 .

25

5^2 means 5×5 .

Skill: Recall 2^2 , 3^2 , 4^2 , 5^2 , 2^3 , and 3^3

3. FOUNDATION 3

Evaluate $10 \times 10 \times 10$.

1000

Three factors of 10 give one thousand.

Skill: Multiply by 10 repeatedly to build powers of 10

4. FOUNDATION 4

How many factors of 2 appear in $2 \times 2 \times 2 \times 2 \times 2$?

5

Count each repeated factor, not the multiplication signs.

Skill: Count how many factors are written

6-3 · Explore Numerical Expressions with Exponents

Connect & apply · 8 tasks · Prepares for 6.EE.A.1

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

5. CHECK A

6^2 ? Write $2 \times 2 \times 2 \times 2$ as a power. 10^3 ?

**$6^2 = 6 \times 6 = 36$. $2 \times 2 \times 2 \times 2$ as a power = 2^4 .
 $10^3 = 10 \times 10 \times 10 = 1,000$.**

Compare each part of your work with the worked solution.

6. CHECK B

7^2 ? Write $3 \times 3 \times 3 \times 3 \times 3$ as a power. 10^4 ?

$7^2 = 49$. $3 \times 3 \times 3 \times 3 \times 3 = 3^5$. $10^4 = 10,000$.

Compare each part of your work with the worked solution.

7. CHECK C

4^3 ? Write $5 \times 5 \times 5$ as a power. 10^2 ?

$4^3 = 64$. $5 \times 5 \times 5 = 5^3$. $10^2 = 100$.

Compare each part of your work with the worked solution.

8. TRANSFER

Which is greater, 3^4 or 4^3 ?

$3^4 = 81$ and $4^3 = 64$, so 3^4 is greater.

Explain why your method works and include units when needed.

6-4 · Write and Evaluate Numerical Expressions with Exponents

Connect & apply · 8 tasks · Prepares for 6.EE.A.2.c

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Evaluate 4^3 .

64

$4 \times 4 \times 4 = 16 \times 4 = 64$.

Skill: Evaluate a power

2. FOUNDATION 2

Evaluate $2 + 3^2 \times 4$.

38

Exponent first gives 9; multiplication gives 36; then add 2.

Skill: Follow order of operations with parentheses and exponents

3. FOUNDATION 3

Evaluate $(8 + 4)/3$.

4

The fraction bar groups the numerator: $12 \div 3 = 4$.

Skill: Read a fraction bar as a grouping symbol

4. FOUNDATION 4

Write an expression for 7 less than the product of 3 and 5.

$3 \times 5 - 7$

Start with the product, then subtract 7. Its value is 8.

Skill: Translate "sum," "product," and "less than" into symbols

6-4 · Write and Evaluate Numerical Expressions with Exponents

Connect & apply · 8 tasks · Prepares for 6.EE.A.2.c

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

5. CHECK A

$4 + 6 \div 2$? $3 + 2^2 \times 5$? Which operation comes first?

$4 + 6 \div 2$: Divide first ($6 \div 2 = 3$), then add: $4 + 3 = 7$. $3 + 2^2 \times 5$: Exponent first ($2^2 = 4$), then multiply ($4 \times 5 = 20$), then add: $3 + 20 = 23$. Exponent comes first, then multiplication, then addition.

Compare each part of your work with the worked solution.

6. CHECK B

$8 + 12 \div 4$? $5 + 3^2 \times 2$? Which operation comes first?

$8 + 12 \div 4$: divide first (3), then add: 11. $5 + 3^2 \times 2$: exponent (9), then multiply (18), then add: 23. The exponent comes first.

Compare each part of your work with the worked solution.

7. CHECK C

$20 - 6 \times 2$? $4 + 2^3 \div 4$? Which operation comes first?

$20 - 6 \times 2$: multiply first (12), then subtract: 8. $4 + 2^3 \div 4$: exponent (8), then divide (2), then add: 6. The exponent comes first.

Compare each part of your work with the worked solution.

8. TRANSFER

Evaluate $2 \times (3 + 4)^2 - 10$.

Parentheses: 7. Exponent: 49. Multiply: 98. Subtract: 88.

Explain why your method works and include units when needed.

6-5 · Write and Evaluate Algebraic Expressions

Connect & apply · 8 tasks · Prepares for 6.EE.A.2

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Evaluate $n + 8$ when $n = 6$.

14

Replace n with 6: $6 + 8 = 14$.

Skill: Substitute a number for a letter, then compute

2. FOUNDATION 2

Evaluate $2x + 3$ when $x = 5$.

13

Substitute, then multiply before adding: $2 \times 5 + 3$.

Skill: Follow order of operations after substituting

3. FOUNDATION 3

Evaluate $4n$ when $n = 7$.

28

A number next to a variable means multiplication:
 4×7 .

Skill: Read $4n$ as $4 \times n$

4. FOUNDATION 4

Write an expression for 9 more than a number n .

$n + 9$

More than indicates adding 9 to the starting number.

Skill: Translate a word phrase into an expression

6-5 · Write and Evaluate Algebraic Expressions

Connect & apply · 8 tasks · Prepares for 6.EE.A.2

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

5. CHECK A

Evaluate $4n + 3$ when $n = 5$. Write an expression for "7 more than a number."

When $n = 5$: $4(5) + 3 = 20 + 3 = 23$. Expression for '7 more than a number': $n + 7$ (or $x + 7$).

Compare each part of your work with the worked solution.

6. CHECK B

Evaluate $3n + 8$ when $n = 6$. Write an expression for "5 less than a number."

$3(6) + 8 = 18 + 8 = 26$. Expression: $n - 5$.

Compare each part of your work with the worked solution.

7. CHECK C

Evaluate $2n - 7$ when $n = 10$. Write an expression for "the product of 6 and a number."

$2(10) - 7 = 20 - 7 = 13$. Expression: $6n$.

Compare each part of your work with the worked solution.

8. TRANSFER

Write an expression for the cost of n tickets at \$12 each plus a \$5 fee, then find the cost for 4 tickets.

$12n + 5$. At $n = 4$: $12(4) + 5 = 53$ dollars.

Explain why your method works and include units when needed.

6-6 · Identify Equivalent Algebraic Expressions

Connect & apply · 8 tasks · Prepares for 6.EE.A.4

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Simplify $5x + 2x$.

7x

Both terms count x units; combine coefficients $5 + 2$.

Skill: Combine like terms such as $5x + 2x$

2. FOUNDATION 2

Evaluate $3x + 4$ when $x = 2$.

10

Substitute 2, then compute $6 + 4$.

Skill: Evaluate an expression at a chosen value

3. FOUNDATION 3

Evaluate $3(4 + 5)$ using distribution.

27

$3 \times 4 + 3 \times 5 = 12 + 15$.

Skill: Apply the distributive property with numbers, e.g. $3(4 + 5)$

4. FOUNDATION 4

Compare $2(x + 3)$ and $2x + 6$ at $x = 4$. Give both values.

14 and 14

$2(4 + 3) = 14$ and $2 \times 4 + 6 = 14$. Matching at one value alone does not prove equivalence for all values.

Skill: Compare two results to decide whether expressions match

6-6 · Identify Equivalent Algebraic Expressions

Connect & apply · 8 tasks · Prepares for 6.EE.A.4

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

5. CHECK A

$5x + 2x = ?$ Evaluate $2(x + 3)$ and $2x + 6$ at $x = 4$
— are they the same?

$5x + 2x = 7x$. At $x = 4$: $2(4 + 3) = 2(7) = 14$; and $2(4) + 6 = 8 + 6 = 14$. Yes, both equal 14 at $x = 4$. The distributive property, $2(x + 3) = 2x + 6$, shows they are equivalent for every x .

Compare each part of your work with the worked solution.

6. CHECK B

$8y + 3y = ?$ Evaluate $3(x + 2)$ and $3x + 6$ at $x = 5$
— are they the same?

$8y + 3y = 11y$. At $x = 5$: $3(7) = 21$ and $15 + 6 = 21$. Yes — the distributive property shows they match for every x .

Compare each part of your work with the worked solution.

7. CHECK C

$9a - 4a = ?$ Evaluate $4(x + 1)$ and $4x + 4$ at $x = 3$
— are they the same?

$9a - 4a = 5a$. At $x = 3$: $4(4) = 16$ and $12 + 4 = 16$. Yes — equivalent for every x .

Compare each part of your work with the worked solution.

8. TRANSFER

Are $3(2x + 4)$ and $6x + 7$ equivalent? Justify.

No. Distributing gives $6x + 12$, not $6x + 7$. At $x = 0$ they give 12 and 7 — different.

Explain why your method works and include units when needed.

6-7 · Find Factors and Multiples

Connect & apply · 8 tasks · Prepares for 6.NS.B.4

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Compute $84 \div 7$.

12

The related fact is $7 \times 12 = 84$.

Skill: Recall multiplication and division facts

2. FOUNDATION 2

List every positive factor of 18.

1, 2, 3, 6, 9, 18

Use factor pairs 1×18 , 2×9 , and 3×6 .

Skill: List every factor pair of a number up to 100

3. FOUNDATION 3

List the first four positive multiples of 7.

7, 14, 21, 28

Multiply 7 by 1, 2, 3, and 4.

Skill: List the first several multiples of a number

4. FOUNDATION 4

Which of 2, 3, 5, 9, and 10 divide 135 evenly?

3, 5, 9

Digit sum 9 makes it divisible by 3 and 9; final digit 5 makes it divisible by 5. It is not even and does not end in 0.

Skill: Test divisibility by 2, 3, 5, 9, and 10

6-7 · Find Factors and Multiples

Connect & apply · 8 tasks · Prepares for 6.NS.B.4

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

5. CHECK A

All the factors of 18? First five multiples of 6?
GCF of 18 and 24?

Factors of 18: 1, 2, 3, 6, 9, 18. First five multiples of 6: 6, 12, 18, 24, 30. GCF of 18 and 24: Common factors are 1, 2, 3, 6; Greatest is 6.

Compare each part of your work with the worked solution.

6. CHECK B

All the factors of 24? First five multiples of 8?
GCF of 24 and 36?

Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24. Multiples of 8: 8, 16, 24, 32, 40. GCF(24, 36) = 12.

Compare each part of your work with the worked solution.

7. CHECK C

All the factors of 20? First five multiples of 5?
GCF of 20 and 30?

Factors of 20: 1, 2, 4, 5, 10, 20. Multiples of 5: 5, 10, 15, 20, 25. GCF(20, 30) = 10.

Compare each part of your work with the worked solution.

8. TRANSFER

You have 24 pencils and 36 erasers to split into identical kits with nothing left over. What is the largest number of kits?

GCF(24, 36) = 12 kits, each with 2 pencils and 3 erasers.

Explain why your method works and include units when needed.

6-8 · Generate Equivalent Expressions

Connect & apply · 8 tasks · Prepares for 6.EE.A.3

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Find the greatest common factor of 18 and 30.

6

Their common factors are 1, 2, 3, and 6; choose the greatest.

Skill: Find the GCF of two numbers

2. FOUNDATION 2

Factor $12x + 20$ using the greatest common factor.

$4(3x + 5)$

Both terms share 4; divide each term by 4 inside the parentheses.

Skill: Apply the distributive property forward and backward

3. FOUNDATION 3

Simplify $4a + 3a + 2$.

$7a + 2$

Combine the a terms; the constant 2 is a different kind of term.

Skill: Combine like terms

4. FOUNDATION 4

Compute $6(10 + 4)$.

84

Distribute: $60 + 24 = 84$.

Skill: Multiply a number by a sum

6-8 · Generate Equivalent Expressions

Connect & apply · 8 tasks · Prepares for 6.EE.A.3

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

5. CHECK A

GCF of 18 and 24? Expand $3(x + 5)$. Factor $4x + 12$.

**GCF(18, 24) = 6. Expand: $3(x + 5) = 3x + 15$.
Factor: $4x + 12 = 4(x + 3)$.**

Compare each part of your work with the worked solution.

6. CHECK B

GCF of 20 and 30? Expand $5(x + 4)$. Factor $6x + 18$.

GCF(20, 30) = 10. $5(x + 4) = 5x + 20$. $6x + 18 = 6(x + 3)$.

Compare each part of your work with the worked solution.

7. CHECK C

GCF of 12 and 16? Expand $4(x + 7)$. Factor $9x + 15$.

GCF(12, 16) = 4. $4(x + 7) = 4x + 28$. $9x + 15 = 3(3x + 5)$.

Compare each part of your work with the worked solution.

8. TRANSFER

Factor $24x + 36$ completely, then check by expanding.

**GCF(24, 36) = 12, so $24x + 36 = 12(2x + 3)$.
Check: $12(2x) + 12(3) = 24x + 36$. ✓**

Explain why your method works and include units when needed.